CUTTING S1-B

VISUAL ASSEMBLY MANUAL

Revision AM-PRE-2 • 2026-10-29 • controlled document

PRE-RELEASE BUILD / QUALIFICATION MANUAL

The design is defined in the released specifications and BOM. Physical fabrication and qualification are NOT complete. Every section carries its status; open gates are registered in section 12 and must close on their named physical tests before production use. Nothing unverified in this document is presented as completed reality.



Machine under construction - reference photo; ratings and dimensions are never taken from photos.

1. Document control

field	value
document	release/Cutting_Assembly_Manual.pdf (the one canonical assembly PDF)
revision / date	AM-PRE-2 / 2026-10-29
source branch	completion/2026-10-29 (pinned base 0f28aa3)
BOM authority	bom/*.csv - purchase/custom lines with dated prices; blanks are not zero cost
spec authority	shared/*.hpp (single active constants source; legacy scales quarantined)
signal contract	requirements/signal-table.md (polarity explicit; unassigned channels fail closed)
firmware	software/machine - host tests green incl. observability/waste/negative-control/recovery groups
analysis	analysis/* manifests (M27/M28/M31/M33/M36 envelope proofs and budgets)
status vocabulary	DEFINED / NOT FABRICATED / OPEN GATE (handoff 14); test completion: BLOCKING_PASS / CLOSED
explicitly excluded	qualification results, delivered-cost claims, safety certification, certified STO

1.1 Revision history

rev	date	change
AM-PRE-1	2026-10-29	initial pre-release manual
AM-PRE-2	2026-10-29	expanded work instructions, vector sheets with dimensions and balloons, handling/waste paths, wiring blocks, commissioning, rigor requirements reflected in qualification section

1.2 How to read this manual

- Each work instruction lists: identity (BOM id), quantity, fastener class/length, torque status, tooling, inspection point, and a warning where a hazard exists.
- Torque values shown as **TORQUE PENDING - RELEASE** are open gates: they must come from the released torque table (fastener class and engagement are given so the value can be derived and recorded at release). Never guess them.
- Vector sheets are schematic views generated from released spec geometry with real coordinates and ballooned BOM identities; they support the instructions but do not replace the native-CAD release gate.
- Where this manual says OPEN GATE, a named record (P-test, TS/TF, or release action) must exist before the status may change.

1.3 Contents

section	content
2	Safety warnings (construction and operation)
3	Machine overview, coordinate contract, station map
4	Build sequence and subassembly index
5	A: base and frame
6	B: cutting axes (x2)
7	C: cutting head and blade tooling
8	D: feed cassettes and measuring modules
9	E: payoff and wire path
10	F: guards; G: cabinet and wiring
11	Handling and waste paths
12	Commissioning and qualification program
13	Maintenance; 14: fault and recovery; 15: spares
16	Open-gate register
17	Photographic and vector sheet index

2. Safety warnings

Read before any construction, power-up or test work. These warnings do not replace the risk assessment and safety validation (P03/P04/P12, OPEN GATE).

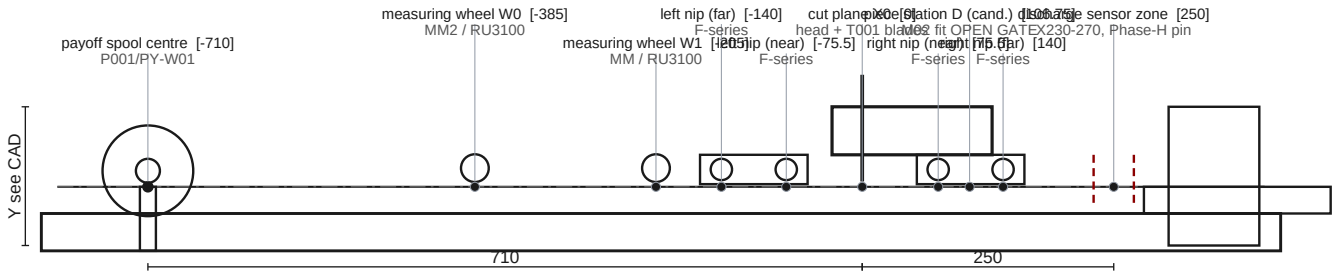
hazard	who is at risk	required control
mains and 36/60 V DC buses	assembler / maintainer	lock out, prove de-energised before cabinet work; stored energy discharged per M21 records
hardened sharp blades, cam/spring motion	assembler / operator	never reach into the head; E-stop removes drive energy but stored energy release must be verified (M37)
payoff spool up to 30 kg plus wire	assembler	two-person lift or lifting aid; the friction brake is NOT a safety brake and guarantees no stop
2500 N-class structural loads	test engineer	proof loads are machine loads, never body loads; exclusion zone during any powered load test
ejected stubs/slugs	operator	waste chute and bin are handling channels, not safety devices; clear only with drives disabled
electrical diagrams herein	assembler	functional reference only; the electrical release (G32) governs conductor sizes, fuses and torque
unqualified machine	all	production use prohibited until section 16 open gates close on signed records

3. Machine overview and station map

Coordinate contract: X = wire travel, Y = opposing cutting strokes, Z = up; cut plane X0; wire at Y0/Z112; +X toward the piece/eject side. Units mm, N, N.m, s. The elevation below shows every real X station used by the sensing/handling contract.

MACHINE SIDE ELEVATION - real X stations (mm from cut plane X0)

schematic assembly view generated from released spec geometry - not a native-CAD render



Key geometry: stock-side measuring wheels W0/W1 (X=-385/-205) can never observe a severed piece; piece motion is observed only at the downstream station D (candidate X=106.75, mechanical fit M02 OPEN GATE); the right cassette cannot propel a severed body shorter than 140 mm; the eject's unobserved remainder is covered by the discharge present->clear evidence (analysis/transport-geometry: 5184-cell envelope proof).

4. Build sequence and subassembly index

order	subassembly	section	depends on	status
A	base and frame	5	-	DEFINED / NOT FABRICATED
B	cutting axes x2	6	A	DEFINED / NOT FABRICATED (G37 load gate)
C	head and blade tooling	7	B	DEFINED / NOT FABRICATED
D	feed cassettes + measuring modules	8	A	DEFINED / NOT FABRICATED (D-station fit gate)
E	payoff and wire path	9	A	DEFINED / NOT FABRICATED
F	guards	10	B,C,D,E	DEFINED / NOT FABRICATED (G34)
G	cabinet and wiring	10	-	DEFINED / electrical release OPEN
H	commissioning + qualification	12	all	OPEN GATE program

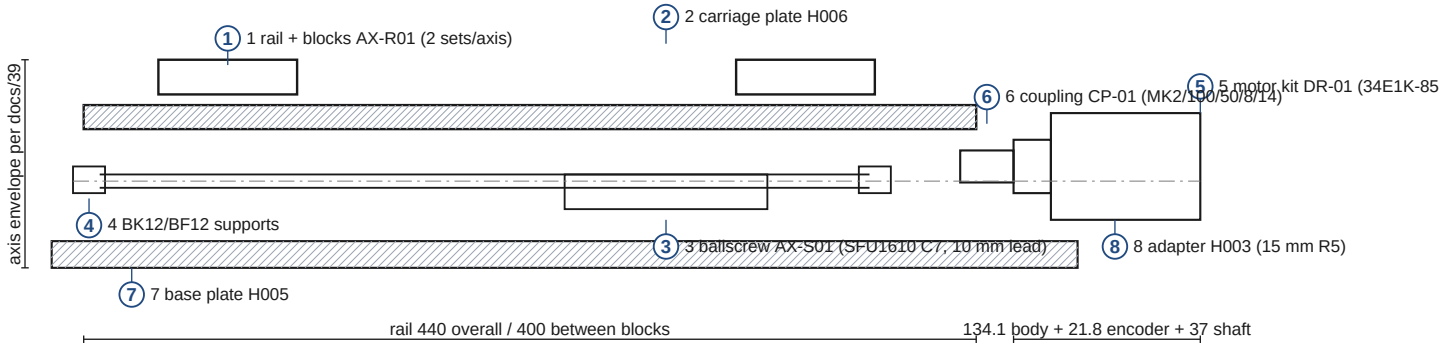
Rule: no subassembly may proceed past its section while any warning or OPEN GATE in that section is unresolved (handoff 14/16.4).

5. A - Base and frame

step	work instruction	fasteners / torque	tools	inspection	status
A-1	Set H001 ground datum plate (180x1040x20) on the prepared floor; reference the cut-plane datum mark.	4x M12 floor anchors (site) - TORQUE PENDING - RELEASE	level, torque wrench	flatness and level record	DEFINED
A-2	Mount H002 risers (110x355x20) to the datum stage grid (46/100).	per drawing - TORQUE PENDING - RELEASE	M8/M10 wrenches	stage grid verified; riser parallelism	DEFINED
A-3	Set the welded frame onto the risers; align to the X-station map (cut plane datum, wheel and cassette stations).	M10x35 class 8.8 bed-to-frame x8 - TORQUE PENDING - RELEASE	straight edge, level	station positions vs section 3 map	DEFINED
A-4	Mount the cabinet plinth and route main cable entries.	per electrical release	electrician tools	entry sealing; G32 open	OPEN GATE

6. B - Cutting axes (2 identical)

AX-01 CUT-AXIS ASSEMBLY - exploded schematic (one of two axes)
balloons map to bom/custom-axis-bom.csv; loads per docs/39 - acceptance-load revalidation OPEN GATE (G37/P05)

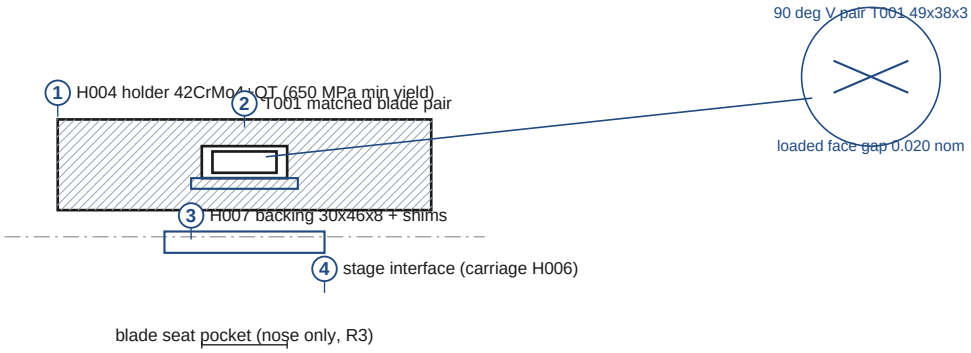


WARNING - proof loading (2500 N axial / 750 N transverse / 375 N vertical, double) is a controlled validation procedure (P05), not an assembly step. The support rating class (4900 N) sits below the 5000 N proof load: this OPEN GATE (G37) is resolved by engineering change, never by rounding.

step	work instruction	fasteners / torque	tools	inspection	status
B-1	Mount H005 base plate (20x86x440) - rail seats milled, 6x 6.6 grid 46/100.	6x ISO4762 M5x16 cl.10.9 class (HF0-B class) + ISO7092 M5 washers - TORQUE PENDING - RELEASE	M5 hex key / torque wrench	plate flatness; seat contact	DEFINED
B-2	Fit AX-R01 rail with 2x HGH20CA blocks (2 sets per axis).	M5 rail screws at 60 pitch - TORQUE PENDING - RELEASE	M5 hex key	rail straightness; block preload class	DEFINED
B-3	Fit AX-S01 ballscrew (SFU1610 C7, 500 mm, 10 mm lead) with DSG16h coupling at the drive end; journal ends 8 h7x18.	nut flange screws per manufacturer - TORQUE PENDING - RELEASE	hex keys, dial indicator	runout; end machining conformance (P01-T3)	DEFINED
B-4	Install BK12/BF12 supports on their lands; align screw to rail.	M5 class screws - TORQUE PENDING - RELEASE	dial indicator	support alignment; screw free rotation	DEFINED
B-5	Fit H006 carriage plate (60x75 pad at 46) to the blocks.	4x ISO4762 M5x16 cl.10.9 (HF0-B) + ISO7092 M5 washers, 8.5 mm nominal engagement - TORQUE PENDING - RELEASE	M5 hex key	49.97 F0 register; PCD70 M5	DEFINED
B-6	Fit motor kit DR-01 (34E1K-85 + CL86T-V41) via H003 adapter (15 mm) and CP-01 coupling (16 mm hub fit on 18 mm exposed shaft).	8x ISO4762 M5x25 cl.10.9 (HM-B) + ISO7092 M5 200HV washers (HM-W, 5.3x9x1, reduced OD 9 mm for the 9.5 mm counterbore); 15 mm adapter thread engagement - TORQUE PENDING - RELEASE	M5 hex key, feeler	134.1 mm rear envelope clearance; coupling alignment (P01-T5)	DEFINED
B-7	Repeat B-1..B-6 for the second axis; label axis identity (blade_front / blade_rear) on the base plate.	-	-	both axes identical	DEFINED

7. C - Cutting head and blade tooling

CUTTING HEAD - assembly front view + blade detail
gap procedure: see work instruction C-4; shim record required (G22)

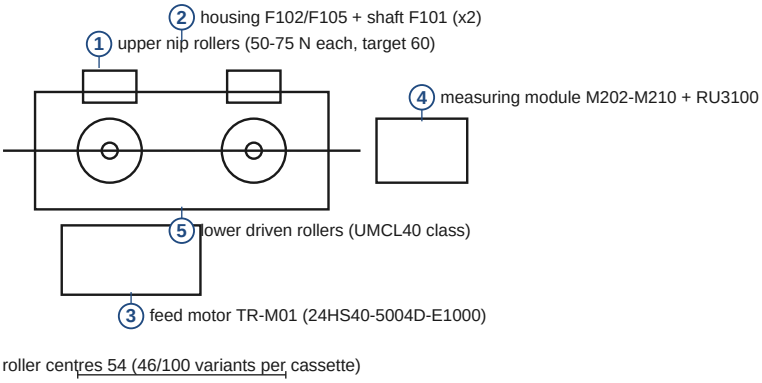


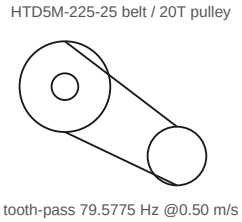
step	work instruction	fasteners / torque	tools	inspection	status
C-1	Mount H004 holder (42CrMo4+QT, 650 MPa min yield) to the carriage; the 70 mm backbone is centred on the screw/load line (R3).	per drawing - TORQUE PENDING - RELEASE	M5/M6 keys	holder centring; nose pocket only machined	DEFINED
C-2	Fit H007 backing plates (30x46x8) and shims; record every shim.	TORQUE PENDING - RELEASE	shim kit, feeler	shim record filed (G22)	DEFINED
C-3	Insert T001 matched 90 deg V blade pair (49x38x3); orientation: V-apex faces the wire centreline.	blade clamps - TORQUE PENDING - RELEASE	microscope 40x	edge radius; apex alignment	DEFINED
C-4	Set loaded blade face gap 0.020 mm nominal at the wire using the shim procedure; execute on both axes.	-	feeler / interferometry	gap record per axis (P07-T6)	OPEN GATE interval
C-5	Record blade serials and cut-count baseline; set the wear inspection plan.	-	-	spares and limit plan (P07-T4)	OPEN GATE

8. D - Feed cassettes and measuring modules

FEED CASSETTE - schematic (one of two: stock side + piece side)

nip forces G23, wheel forces G24 measured per P06; belt tension 180-200 N both stations



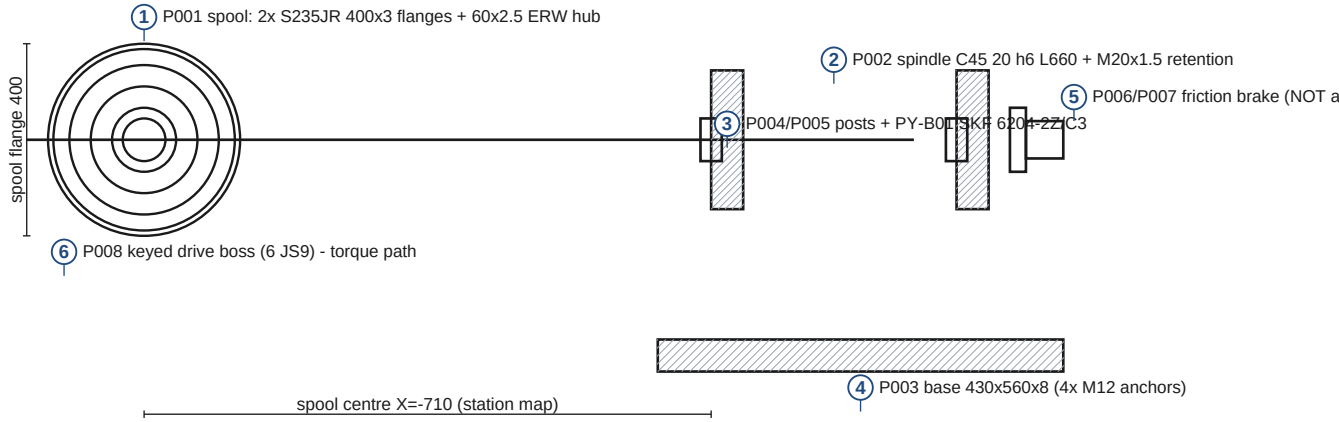


step	work instruction	fasteners / torque	tools	inspection	status
D-1	Assemble F-series cassette (housing F102/F105, shaft F101 x2, rollers, springs; 46/100 or 64/47 centres per variant).	TR-FB01 ISO4762 M5x16 cl.10.9 x8 (roller endplate); TR-FB02 ISO4762 M4x20 cl.10.9 x16 + TR-FW02 ISO7092 M4 washers (F112 front module); TR-FD02 ISO8734 3m6x20 dowels x16; TR-FB03 ISO4762 M3x8 cl.8.8 x32 (F103 cover) - TORQUE PENDING - RELEASE	hex keys, press	roller free rotation; endplate parallelism	DEFINED
D-2	Fit feed motor TR-M01 (24HS40-5004D-E1000, 10 mm shaft) and the HTD5M belt with 20T pulley; set tension 180-200 N (target 190) at BOTH stations.	pulley clamp screws - TORQUE PENDING - RELEASE	tension meter	both station tensions recorded (P06-T3)	DEFINED
D-3	Mount measuring wheel modules W0/W1 (ifm E60137, 63.66 mm OD) with RU3100 encoders on the stock side (X=-385/-205).	ME-FB01 ISO4762 M4x12 cl.8.8 x6 (M201 encoder plate; ifm M4x0.7x6 taps, 6.0 mm nominal engagement; encoder thread torque limit PENDING)	M4 hex key	wheel force 6-19 N each (P06-T2)	DEFINED
D-4	Downstream piece station D (candidate X=106.75): do NOT install until the M02 fit study closes (63.66 mm wheel/encoder/housing among the nips at 75.5/138).	-	-	M02 fit gate; Phase-H pin assignment (analysis/phase-h-io)	OPEN GATE
D-5	Route encoder cabling to GP8-11; leave station-D and handling channels unassigned until Phase H (they fail closed in firmware).	-	cable ties	signal-table conformance (P04-T4)	OPEN GATE

9. E - Payoff and wire path

PAYOFF - exploded schematic (docs/26 R2)

<=30 kg spool + wire; two-person lift; keyed boss P008 required for brake torque path



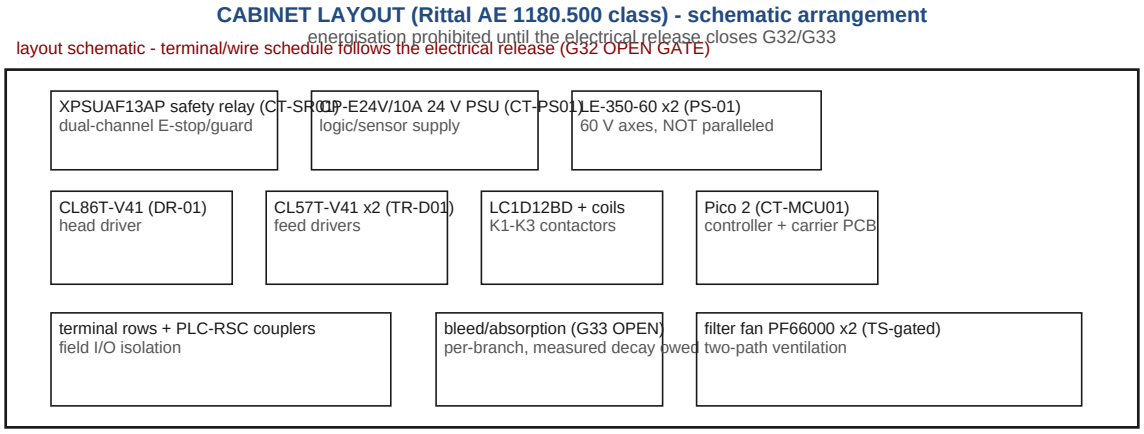
step	work instruction	fasteners / torque	tools	inspection	status
E-1	Set P003 base (430x560x8) level; anchor with 4x M12.	4x M12 anchors - TORQUE PENDING - RELEASE	level, wrench	base level record	DEFINED
E-2	Fit P004/P005 support posts with SKF 6204-2Z/C3 bearings in the 47 H7 pockets (14 deep).	M8 base bolts - TORQUE PENDING - RELEASE	bearing press tool	pocket H7; bearing seating (P01-T1 class)	DEFINED
E-3	Insert P002 spindle (C45 20 h6 L660); fit M20x1.5 retention nuts with tab washers.	M20x1.5 KM-class nuts (PY-R01) - TORQUE PENDING - RELEASE	pin wrench	spindle free rotation; axial float limit	DEFINED
E-4	Fit P008 drive boss (keyseat 6 JS9) welded to the LEFT flange inside the hub tube BEFORE spool assembly.	weld per fabrication procedure	welding shop	weld inspection; key fit	DEFINED
E-5	Assemble P001 spool (2x S235JR 400x3 flanges + 60x2.5 ERW hub 160) and mount with P006 disc + P007 caliper brake.	set screws / caliper adjuster M8 DIN933 - TORQUE PENDING - RELEASE	hex keys	brake effective ONLY with P008 fitted	DEFINED
E-6	Weigh the full spool per lot (wire included); verify <=30 kg and record mass.	-	scale	lot mass record (docs/26 s4); two-person lift	DEFINED

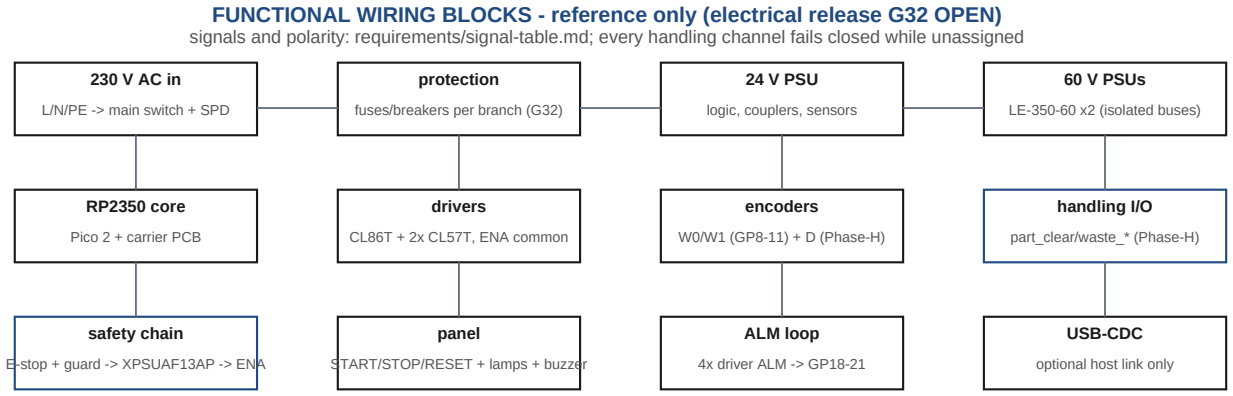
10. F - Guards; G - Cabinet and wiring

F: Guards

step	work instruction	fasteners / torque	inspection	status
F-1	Fit guards over the head, cassettes, payoff and takeaway per the layout; include reel/buffer/conveyor hazards.	per drawing - TORQUE PENDING - RELEASE	opening/distance check vs reachable hazards (P04-T2)	OPEN GATE G34
F-2	Fit guard switches (SLC-F-024-20/20-R4 class) and route to the XPSUAF13AP dual channel.	TORQUE PENDING - RELEASE	dual-channel validation (P04-T1)	OPEN GATE G35

G: Cabinet and wiring





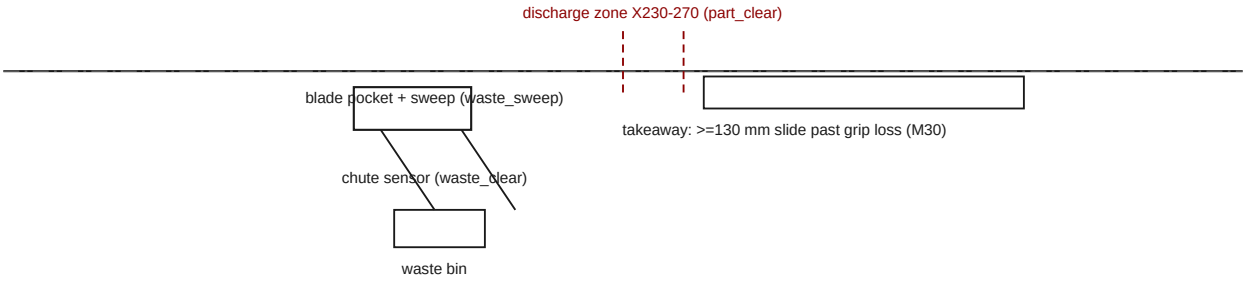
step	work instruction	inspection	status
G-1	Build the cabinet per the layout: 230 V in -> protection -> 24 V PSU -> isolated 60 V PSUs -> drivers -> core. No bus paralleling.	point-to-point against requirements/signal-table.md	OPEN GATE G32
G-2	Safety chain: E-stop (SIRIUS ACT 3SU1851) + guard loop -> XPSUAF13AP -> ENA common (GP17). Dual channel all the way.	single-fault tolerance test (P04-T1)	OPEN GATE G35
G-3	Field inputs via PLC-RSC 24DC/21 couplers; wetting current must exceed relay minima (the old 57.9 uA defect class must not recur).	wetting measurement (P03-T5)	DEFINED
G-4	ALM lines to GP18-21; encoders to GP8-11 via AM26LV32E receivers; USB-CDC is optional and never required for motion.	continuity + polarity audit (P04-T4)	DEFINED
G-5	Leave part_clear / waste_clear / waste_sweep / station-D unassigned until Phase H; firmware fails closed meanwhile.	verify fail-closed behaviour on power-up	OPEN GATE Phase-H

11. Handling and waste paths

HANDLING AND WASTE PATHS - body classes and required evidence

acceptance needs discharge present->clear in window; waste needs chute transit; bodies <140 mm are never propel-lable

200-3000 mm part	grip to X140 -> coast -> takeaway -> clear
3-25 mm slugs	release at open -> sweep -> chute
10 mm prime stub	sweep -> chute



The three body classes have different, mandatory evidence chains (firmware enforces them in production):

- **Prime stub (10 mm):** severed at the plane; it cannot reach the nearest right nip (75.5 mm) and is never propelled by feed motion. The constrained sweep (waste_sweep) must produce a chute transit (waste_clear present->clear) before any further feed or blade close.
- **Strip slugs (3-25 mm):** released at blade open after each score+separation; same sweep and chute-transit requirement at LeadOpen/TailOpen.
- **Finished part (200-3000 mm):** driven by the right cassette only while its tail is inside the far nip (X140); then the takeaway (>=130 mm slide) clears the discharge zone X230-270. Acceptance requires the discharge present->clear pair inside the ejection observation window - never commanded steps alone.
- **Interrupted in-flight body:** classified by real length at the prime re-cut - short goes to the sweep, long is scrap-discharged through the takeaway and counted scrap (recovery drill P08-T6).

Blocked sweep, stuck sensors and early/late clears all fault and quarantine - they never produce accepted counts (firmware regression groups waste / observability / recovery).

12. Commissioning and qualification program

12.1 Commissioning sequence

1. Flash the released firmware image; record its SHA-256 in the machine log. The factory test runs from a separate UF2 and must exercise every I/O channel.
2. Power up with drives disabled; verify fail-closed handling channels report NOT clear.
3. Home both blade axes; route wire through both cassettes (threading per section 3).
4. Service fixture mode (`starz-ctl ... fixture on`) is for engineering runs only: it never emits accepted counts or persistent approval. Leave it OFF for production.
5. Run the commissioning polarity moves for all four axes (station D when fitted); calibrate W0/W1/D (starz-ctl calib after the length record).
6. Approve the retained recipe over the host link (Msg::Approve) - approval is keyed to calibration + recipe + firmware identity and is cleared by any change.
7. Production START requires the approval; start with commissioning speed (0.05 m/s class) and the handling verification ON (default).
8. Run the qualification program below; only signed records change any status.

12.2 Qualification program (P00-P15 families)

family	scope	records
P00 receiving	identity, first article, evidence/price reconciliation	tests/P00-P15/records/P00-T*.json
P01 fits	interference pairs, runout, torques, alignment	P01-T*.json
P02/P03/P04 electrical	protection curves, power components, safety wiring	P02-T*..P04-T*.json
P05 loads	proof loads and post-proof metrology	P05-T*.json
P06 transport	nip/wheel forces, belt tension, loaded radius	P06-T*.json
P07 quality	cut/strip coupons, blade life, face gap	P07-T*.json
P08 handling	stub/slug/output evidence chains	P08-T*.json
P09 dynamics	tooth-pass, Campbell, FRF, damping	P09-T*.json
P10 thermal	18 TS steady + 17 TF fault (tests/P10-thermal/)	TS*.json / TF*.json
P11 electrical system	regen decay, bleed timing, long-run behaviour	P11-T*.json
P12 safety validation	PLr/SIL, ISO 13849-2, engineer sign-off	P12-T*.json
P13 accuracy	length/strip capability, gauge R&R;, budgets	P13-T*.json
P14 endurance	duty blocks, wear limits, service intervals	P14-T*.json
P15 integration	independent build, collisions, final inspection	P15-T*.json

All records ship executed=false. Rigor requirements (traceability, |e|+U<=T conformity, sample sizes, predeclared analysis) are stated in tests/P00-P15/procedures.md and apply to every test.

13. Maintenance (provisional until P14/TS evidence)

item	interval (provisional)	limit / action	record
ventilation filters (2 paths)	inspect each shift; replace per TS evidence	installed flow below service limit triggers change	P10 + TS01-TS18
blades T001	inspect each shift; replace per P07-T4 limit	edge radius/wear limit from coupons	P07-T4
measuring wheels W0/W1/D	weekly	tread wear and calibration drift check	P06-T2/T5
nips and guides	weekly	50-75 N nip force check	P06-T1
belt tension	monthly + after any drive work	180-200 N at both stations	P06-T3
payoff brake pads	monthly	friction pad wear	P14-T2
spindle bearings	per P14 duty evidence	lubrication interval to be established	P14-T3
safety functions	per P12 proof-test interval	E-stop/guard validation repeat	P12-T4

14. Fault and recovery (operator)

fault	meaning	operator action	do NOT
red LED steady	latched fault; drives disabled	clear cause; hold RESET 0.5 s; rehome	force motion without rehoming
no_wire / length_slip	material motion mismatch	inspect path and wheels; rethread; RESET	accept the piece in the machine
part_not_cleared	discharge evidence missing (quarantine)	remove/verify part at the takeaway; log scrap	count the piece as accepted
waste_not_cleared	stub/slug did not transit the chute	clear sweep chute and pocket; RESET	feed over a trapped body
wire_geometry	planned move/score rejected by the wire model	rethread and rehome; report the case	bypass the check
power loss mid-piece	in-flight part uncertain	on restart: rehome + rethread; prime scrap-discharges leftovers	resume as if coordinates were truth

15. Spare parts (initial)

part	identity	qty	status
blades	T001 matched 90 deg V pair 49x38x3	2 pairs	DEFINED / NOT FABRICATED
filters	Pfannenberg 11666102055 / 11760002055 candidates	1 set	OPEN GATE TS evidence
measuring wheels	ifm E60137	1	DEFINED
encoders	ifm RU3100	1	DEFINED
belt	HTBN225S5M-250	1	DEFINED
couplings	R+W MK2/100/50/8/14	1	DEFINED
fuses/relays	per electrical release	-	OPEN GATE G32
brake pads	docs/26 P007 friction pads	1 set	DEFINED

16. Open-gate register (production prohibited until closed)

gate	what is owed	record / owner
G37 axis loads	revalidate acceptance loads vs the frozen 10 mm-lead screw; resolve 4900 N < 5000 N proof contradiction by change	P05-T6
G16 fits	3-6 um interference matched pairs + bearing clearance envelope	P01-T1 / P13
G18 shaft thermal	shaft rise <=20 K above housing (stored 29.66 K case fails)	P10
G22 face gap	0.020 mm loaded interval derived from cut-quality data	P07-T6
G29-G30 ventilation	installed filtered flow + thermal margins; 18 TS + 17 TF	P10 program
G32 protection	exact fuse/breaker coordination; conductor/terminal limits	P02 / electrical release
G33 regen/bleed	per-branch absorption/bleed with measured bus decay	P11-T1/T2
G35 safety	risk-derived PLr/SIL; dual-channel validation; competent engineer	P12 sign-off
M02 D-station fit	wheel/encoder/housing fit in X38-140 among nips 75.5/138	fit study + P01
Phase-H I/O	assign part_clear / waste_clear / waste_sweep / station-D pins	analysis/phase-h-io option study
G39 RFQs	send the drafted enquiries (7 exact RFQs)	procurement/rfq-drafts.md
G47 cost	complete landed cost; 70 % like-for-like comparison	analysis/cost-basis (NOT ESTABLISHED)
P15 independence	independent builder completes this manual without choices	P15-T1

17. Photographic and vector sheet index

Reference photos from the machine under construction. They establish no ratings, dimensions or qualification.



rear view (reference)



payoff and buffer (reference)

Vector sheets (drawn from released spec geometry; real coordinates):

- sheet 1 - machine side elevation with the station dimension chain (section 3)
- sheet 2 - AX-01 cut-axis exploded schematic (section 6)
- sheet 3 - cutting head front view + blade detail (section 7)
- sheet 4 - feed cassette schematic + belt side view (section 8)
- sheet 5 - payoff exploded schematic (section 9)
- sheet 6 - cabinet layout and wiring blocks (section 10)
- sheet 7 - handling and waste paths (section 11)

End of manual. Status changes require signed records referenced in section 16.